DURGESH UNDE

AI ENGINEER | LLM APPLICATIONS, AGENT SYSTEMS & EVALUATION

Shrirampur, Maharashtra, India | +91 93223 23097 | durgeshunde@gmail.com
github.com/durgeshunde | Engineering portfolio | LinkedIn

AI engineer focused on LLM adaptation, agent systems and evaluation. Built a native desktop AI application, a serverless conversational assistant and a reproducible multi-agent benchmark spanning 630 LLM trajectories. Founder of XYVERION AI with professional model evaluation experience.

TECHNICAL SKILLS

LLM engineering: PyTorch, Hugging Face Transformers, PEFT, LoRA/QLoRA, bitsandbytes, NF4 quantization

Agent systems: Tool calling, model APIs, conversation memory, desktop automation, SQLite, Win32 APIs

Development: Python, JavaScript, React, HTML/CSS, REST APIs, Git, Cloudflare Workers, KV storage

Evaluation: Controlled experiments, threshold calibration, mutual information, grounding, human feedback

SELECTED ENGINEERING & RESEARCH

When Correlation Lies | Multi-Agent Evaluation | Apart Sprint, Oct 2026

- Built a four-agent Python benchmark to distinguish shared-context correlation from supplied private coordination; evaluated Qwen2.5-1.5B-Instruct across 630 trajectories at 5, 20 and 50 rounds.
- Implemented Pairwise Net Influence, mutual information and conditional MI detectors, with calibration and held-out evaluation to examine false alarms and sensitivity to trajectory length.
- Separated trajectory logs from ground-truth labels and documented reproduction commands. Code & submitted report.

XYVERION 3.0 | Local LLM & Desktop Agent System

- Connected local language model inference to native Windows controls through Python ctypes and Win32 APIs, with SQLite conversation memory and an Edge WebView2 interface.
- Built a model adaptation pipeline using PyTorch, Transformers, PEFT and 4-bit QLoRA, covering instruction data preparation, validation loss tracking and model weight fusion.
- Implemented GPU memory lifecycle handling to support model loading and inference on constrained hardware.

XYVERION AI | Conversational Web Application

- Built a serverless assistant with Cloudflare Workers, model API orchestration and KV-backed conversation state.
- Added bounded retries, model fallback handling and email notifications around the conversation workflow. Live demo.

Selected technical writing: The Evolution of Generative AI Models (2025), an independent review of transformers, multimodal systems, agent workflows and memory.

PROFESSIONAL EXPERIENCE

Founder & AI Engineer | XYVERION AI | Sep 2026 - Present

- Own engineering across model adaptation, agentic workflows, API integration and application interfaces; translate use cases into working desktop and web AI demonstrations.

AI Data Annotation & Evaluation Contractor | Innodata India Pvt. Ltd. | Aug 2026 - Present

- Evaluate AI outputs for accuracy, relevance and grounding; perform reference-expression annotation, linguistic review and training data quality assurance.

AI Training & Data Annotation Contributor | Outlier AI | Jan 2026 - Jul 2026

- Provided human feedback, model output evaluation and multimodal annotation for AI training projects.

SELECTED CERTIFICATIONS & TRAINING

AWS: Databases for Agentic AI (microcredential), Oct 2026

Anthropic: Model Context Protocol: Advanced Topics; Claude with Amazon Bedrock, Oct 2026

Dataiku: Generative AI Practitioner, Oct 2026

EDUCATION & LANGUAGES

University of the People: B.S. in Computer Science (pursuing)

HSC, Science: 78.50%, 2022 | **IELTS Academic:** 7.5 overall, CEFR C1, September 2025

Languages: English, Hindi, Marathi and German